

$$T(n) = 3T\left(\frac{n}{2}\right) + \boxed{\cancel{c \cdot n^2 \cdot \sqrt{n}}} \ominus (n^2 \sqrt{n})$$

$$n^{\log_2 3} \parallel 4n^2 \cdot \sqrt{n}$$

$$n^{1.5} \quad 4n^2 n^{1/2} = 4n^{2.5} \quad \boxed{\text{CASO 3}}$$

$$f(n) = \Omega(n^{1.5+\epsilon})$$

$$a \quad f\left(\frac{n}{b}\right) \leq c \cdot f(n) \quad \boxed{0 < c < 1}$$

$$3 \left(\overset{\downarrow (\frac{n}{2})^2}{\frac{n^2}{2^2}} \cdot \overset{\downarrow \sqrt{\frac{n}{2}}}{\frac{\sqrt{n}}{\sqrt{2}}} \right) \leq c \cdot n^2 \cdot \sqrt{n}$$

$$\frac{3}{\sqrt{2}} \cancel{n^2 \cdot \sqrt{n}} \leq c \cdot \cancel{n^2 \cdot \sqrt{n}}$$

$$\boxed{\frac{3}{4 \cdot \sqrt{2}}} \leq c \quad c = \frac{3}{\sqrt{2}} > 1$$

$$\frac{3}{2^{2.5}} \leq c$$

$$\boxed{c = \frac{3}{2^{2.5}}}$$

$$T(n) = 10T\left(\frac{n}{3}\right) + 3n^2 + n$$

$$n^{\log_3 10}$$

||

$$\cancel{3n^2 + n}$$

$$2 < \log_3 10 < 3$$

CASO 1

$$\Theta(n^{\log_3 10})$$

$$f(n) = O(n^{\log_3 10 - \epsilon})$$

Algo(A, n)

{ if $n < 1$ then return 1

$n \geq 1$
 $t > n^2$

→ $t = \text{Algo}(A, \underline{n/2})$

input di dimensione $n/2$

→ if $t > n^2$ then

$t = t - \frac{1}{2} \text{Algo}(A, \underline{n/2})$

→ for $j = 1$ to n

$t = t + A[j] + \text{Algo1}(n)$

} return t

$\hookrightarrow n \cdot \sqrt{n}$

$\text{Algo1}(n)$ ha una complessità

$\Theta(\sqrt{n})$

$$T(n) = 2T\left(\frac{n}{2}\right) + n \cdot \sqrt{n}$$

$n^{\log_2 2}$

//

$n^{1.5}$

n

$n^{1.5}$

$$\rightarrow f(n) = \Omega(n^{\log_b a + \epsilon}) \quad \underline{\underline{\text{CASO 3}}}$$

$$n^{1.5} = \Omega(n^{1+\epsilon})$$

$$a \cdot f\left(\frac{n}{b}\right) \leq c \cdot f(n)$$

$$\cancel{2} \cdot \cancel{\frac{n}{2}} \cdot \frac{\sqrt{n}}{\sqrt{2}} \leq c \cdot \cancel{n} \cdot \cancel{\sqrt{n}}$$

$$\Theta(n\sqrt{n})$$

$$\frac{1}{\sqrt{2}} \leq c$$

$$c = \frac{1}{\sqrt{2}}$$

$Algo(A, i, j)$ $Algo(A, \underline{1}, m)$

{ if $(j < i)$ then return $\emptyset$

else if $i = j$ then return $A[i]$

else {

$d = (j - i + 1) / 3$

return $Algo(A, i, i + d - 1) +$

$Algo(A, i + d, i + 2d - 1) +$

$Algo(A, i + 2d, j)$

}

$$T(m) = 3T(m/3) + C$$

$$m^{\log_3 3}$$

$$// C$$

$$\underline{m}$$

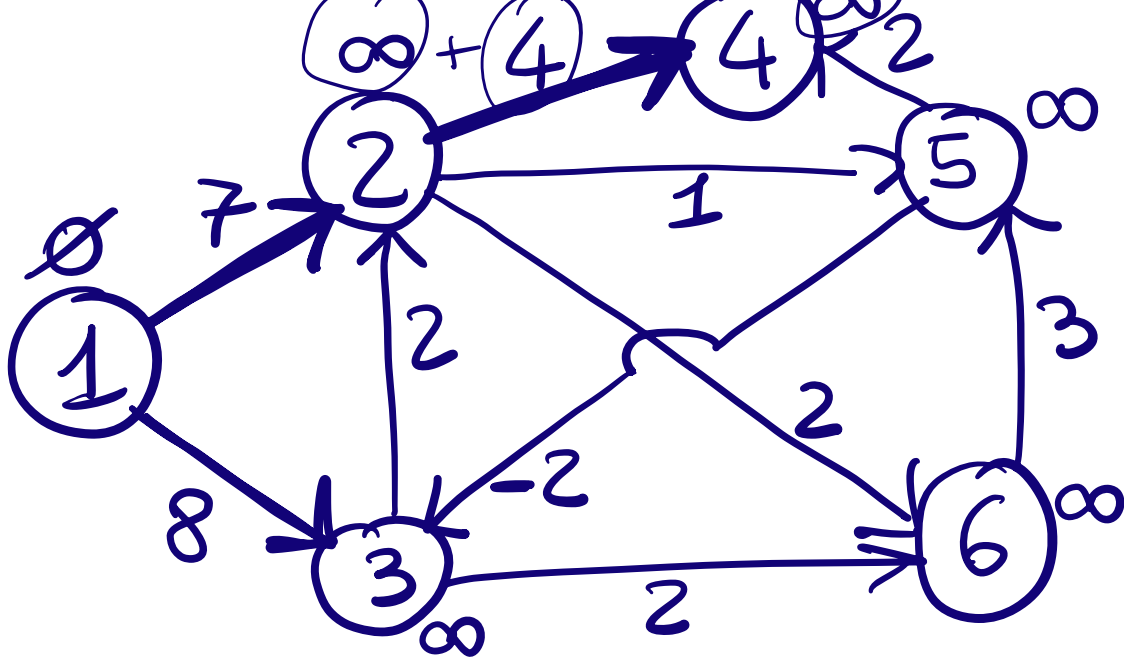
$$// C$$

CASE 1

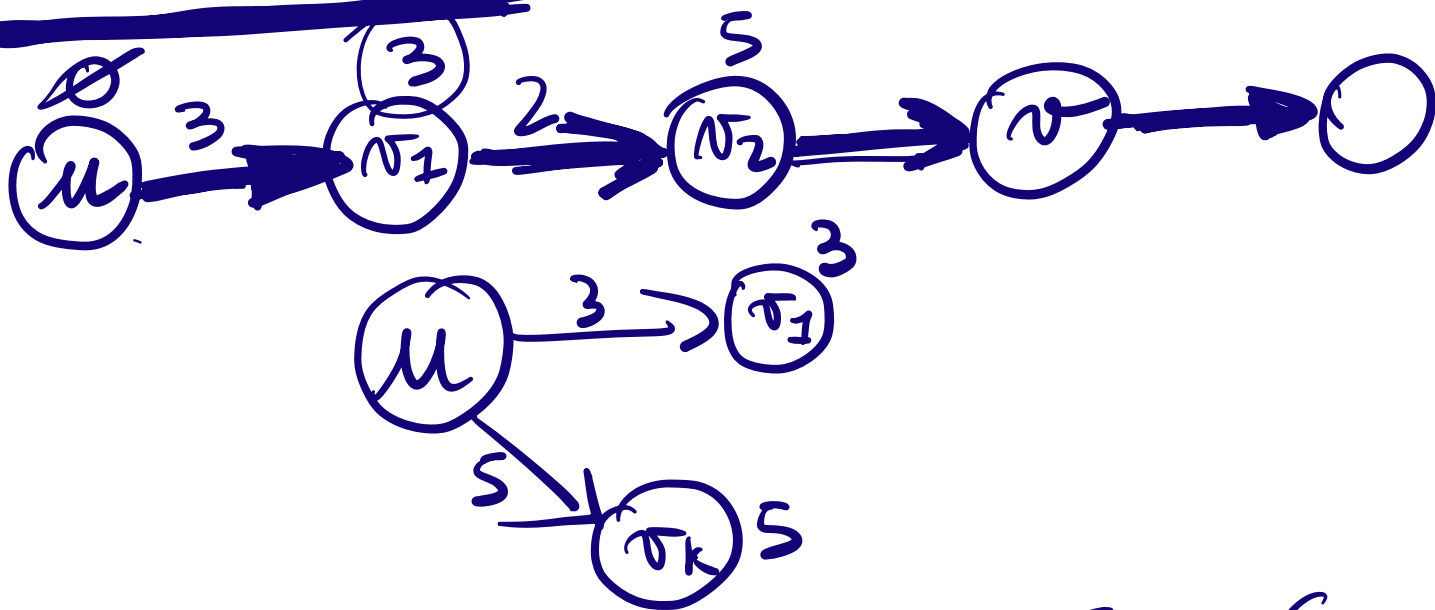
$$C = O(m^{1-\epsilon})$$

$$\Theta(m)$$

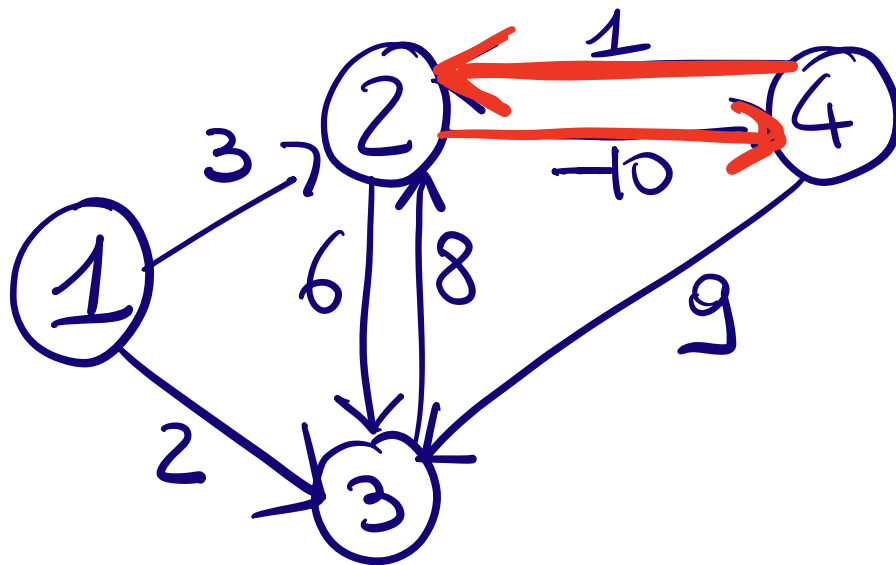
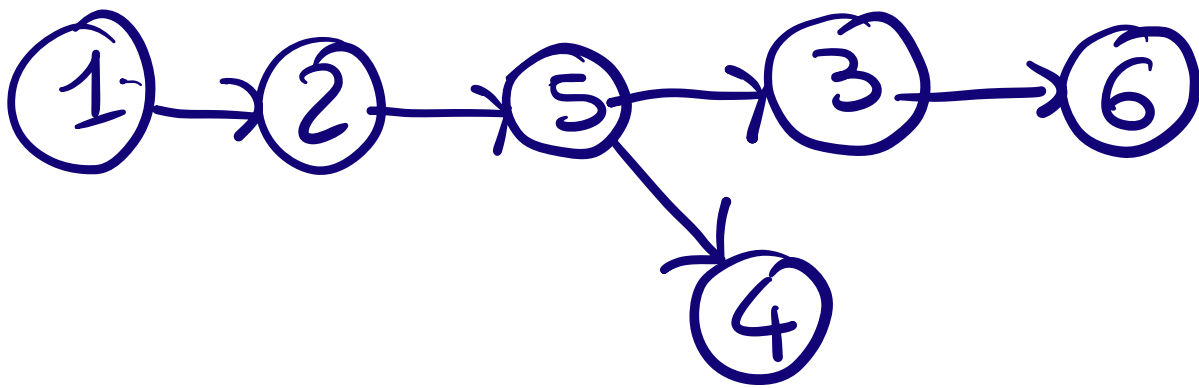
$$\underline{\Theta(m^{\log_3 3})} \subseteq \Theta(m)$$



CAMMINI MINIMI DAL NODO 1 CHE HANNO AL
PIU' 4 ARCHI.



	1	2	3	4	5	6
$i=\emptyset$	$\emptyset, \infty$	∞, ∞	∞, ∞	∞, ∞	∞, ∞	∞, ∞
$i=1$	$\emptyset, \infty$	7, 1	8, 1	∞, ∞	∞, ∞	∞, ∞
$i=2$	$\emptyset, \infty$	7, 1	8, 1	11, 2	8, 2	9, 2
$i=3$	$\emptyset, \infty$	7, 1	6, 5	10, 5	8, 2	9, 2
$i=4$	$\emptyset, \infty$	7, 1	6, 5	10, 5	8, 2	8, 3



Floyd-Warshall

$D^{(0)}$

	1	2	3	4
1	0	3	2	∞
2	∞	0	6	-10
3	∞	8	0	∞
4	∞	1	9	0

$D^{(1)}$

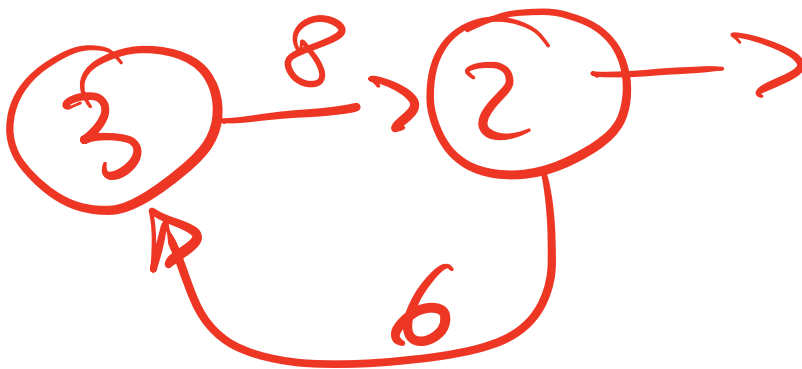
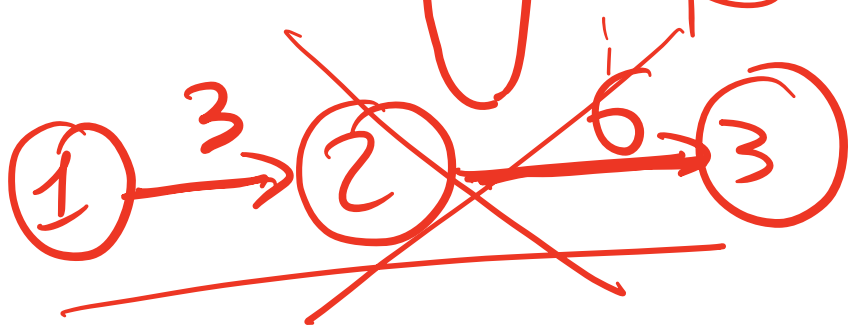
	1	2	3	4
1	N	1	1	N
2	N	N	2	2
3	N	3	N	N
4	N	4	4	N

$D^{(1)}$

	1	2	3	4
1	0	3	2	-7
2	∞	0	6	-10
3	∞	8	0	-2
4	∞	1	7	-9

$\Pi^{(1)}$

	1	2	3	4
1	N	1	1	2
2	1	N	2	2
3	1	3	N	2
4	1	4	2	N



$D^{(2)} =$

	1	2	3	4
1	0	3	2 -7	
2	∞	0	6	-10
3	∞	8	0	-2
4	∞	1	7	0

$\Pi =$

	1	2	3	4
1	N	1	1	2
2	1	7	2	2
3	7	3	N	2
4	7	4	2	N